Linear Independence And Spanning

Linearly Dependence/Independence

Imagine you're a cybersecurity analyst at a tech company. Your AI system uses 3 algorithms to detect intrusions

- **Algorithm A:** Detects brute-force attacks (vector $\mathbf{v}_1$).
- **Algorithm B:** Flags unusual login times (vector $\mathbf{v}_2$).
- **Algorithm C:** A 'mystery' algorithm (vector $\mathbf{v}_3$) that's just $\mathbf{v}_3 = 2\mathbf{v}_1 + \mathbf{v}_2$.

One night, hackers breach your system, but your AI fails to catch them. Why?

Because Algorithm C was a **fake safeguard**—it didn't add new protection! It was just a **linear combination** of A and B. This is the disaster of **linear dependence**!"

Definition

Linearly Dependent Vectors:

When one algorithm (vector) is just a 'remix' of others (e.g., $v_3 = 2v_1 + v_2 \Rightarrow 2v_1 + v_2 - v_3 = 0$*) your system has* ***redundant*** *defenses (IT Analogy).*

Mathematical Def: A set of vectors $\{\overrightarrow{v_1}, \overrightarrow{v_2}, \dots, \overrightarrow{v_n}\}$ is **linearly dependent** if:

$$a_1\overrightarrow{v_1} + a_2\overrightarrow{v_2} + \dots + a_n\overrightarrow{v_n} = 0$$

for some **not all zero** scalars $a_1, a_2, \dots, a_n$.

This means **one or more vectors** in the set can be written as a **linear combination of the others**.

Linearly Independent Vectors:

When each algorithm adds ***unique*** *protection (no remixes), your security is robust. (IT Analogy)*

Mathematical Def: A set of vectors $\{\overrightarrow{v_1}, \overrightarrow{v_2}, \dots, \overrightarrow{v_n}\}$ is **linearly independent** if:

$$a_1\overrightarrow{v_1} + a_2\overrightarrow{v_2} + \dots + a_n\overrightarrow{v_n} = 0 \Rightarrow a_1 = a_2 = \cdots = a_n = 0$$

No vector in the set can be written as a combination of the others — each vector carries **unique** information.

When Linear Dependence is *Bad* (Redundancy = Waste)

Example 1: Cryptography (Weak Keys)

Scenario: Encryption keys $k_1, k_2, and\ k_3 = k_1 + k_2$

Risk: Hackers only need to crack k_1 and k_2 to derive k_3

Solution: Use **independent** keys.

Example 2: Machine Learning (Overfitting)

Scenario: Your dataset has two features:

RAM_usage and RAM_usage_percent

Problem: These features are **linearly dependent as** RAM_percent = k × RAM_usage

Disaster: Your model wastes time on redundant calculations, risking overfitting.

Fix: Drop one feature.

When Linear Dependence is *Useful* (Efficiency = Smart Design)

Example 1: Error Correction (RAID Storage)

Scenario: RAID-5 uses **parity bits** (linear combinations of data bits).

Benefit: If one disk fails, data can be **reconstructed** from dependent parity bits.

Example 2: Computer Graphics (Baking Lighting)

Scenario: In game engines, static light calculations

$$(Light = a \times Diffuse + b \times Specular)$$

are **precomputed** (baked) into textures.

Why? Saves real-time computation by reusing dependent values.

The Key Insight

Dependence = Redundancy → *Bad* when it wastes resources (e.g., ML features, crypto).

Dependence = Structure → *Good* when it enables optimization (e.g., compression, error recovery).

A password manager stores password3 = Password 1 + Password 2

Bad (security risk!).

A video codec stores Frame 2 = Frame 1 + Motion Diff

Good (efficient compression!)

Example:

You are developing a **gesture recognition system** for a VR (Virtual Reality) game. The system analyzes hand movements in 3D space and represents each motion pattern as a **motion vector**:

$\vec{u} = (1, -2, 3)$: Swipe Left and $\vec{v} = (5, 6, -1)$: Swipe up and $\vec{w} = (3, 2, 1)$: Swipe Forward

These motion vectors are stored as **feature vectors** in your classification model.

Now, as a smart developer, you want to **optimize** your model by checking if any of these features are **redundant** (i.e., one can be formed from others).

Question: Are the above vectors **linearly dependent** or **independent**?

To determine linear dependence, check if: $a\vec{u} + b\vec{v} + c\vec{w} = 0$

$$a(1, -2, 3) + b(5, 6, -1) + c(3, 2, 1) = (0, 0, 0)$$

$$a + 5b + 3c = 0, \qquad -2a + 6b + 2c = 0, \qquad 3a - b + c = 0$$

Solving the system by Gauss Jordan elimination method, we have $a = 11b, \quad c = -2b$

So, the vectors are linearly dependent.

Example:

Are the vectors $P = t^2 + t + 2$, $Q = 2t^2 + t + 1$ and $R = 3t^2 + 2t + 2$ in P_2 are linearly independent?

Solution: Let $\boldsymbol{k_1 v_1 + k_2 v_2 + k_3 v_3} = 0$

$$k_1(t^2 + t + 2) + k_2(2t^2 + t + 1) + k_3(3t^2 + 2t + 2) = 0t^2 + 0\mathrm{t} + 0$$

$$k_1 t^2 + k_1 t + 2k_1 + 2k_2 t^2 + k_2 t + k_2 + 3k_3 t^2 + 2k_3 t + 2k_3 = 0t^2 + 0\mathrm{t} + 0$$

$$t^2(k_1 + 2k_2 + 3k_3) + t(k_1 + k_2 + 2k_3) + (2k_1 + k_2 + 2k_3) = 0t^2 + 0\mathrm{t} + 0$$

$$k_1 + 2k_2 + 3k_3 = 0, \qquad k_1 + k_2 + 2k_3 = 0, \qquad 2k_1 + k_2 + 2k_3 = 0$$

Solving the system by Gauss Jordan elimination method, we have $k_1 = 0$, $k_2 = 0, k_3 = 0$

Hence the given polynomial vectors are linearly independent.

Spanning Set

Imagine you are developing a joystick controller for a 3D drone simulator. Joystick can move the drone in certain directions… like forward, sideways, or diagonally.

But here's the question: Can joystick system **reach any position in 3D space** using a combination of available moves?

That's what the idea of a **spanning set** is! If your available motion vectors (basic joystick directions) can be combined to reach **any point in space**, then they **span the space**.

A set of vectors **spans** a space if their linear combinations can **generate every vector in that space**.

Spanning Set = A toolbox. If tools can build anything in space, toolbox is complete.

Definition

Mathematical Def: A set of vectors $\{\overrightarrow{v_1}, \overrightarrow{v_2}, \dots, \overrightarrow{v_n}\}$ in a vector space $\underline{V}$ is said to **span *V*** if:

every vector $\underline{v} \in V$ can be written as a linear combination of $\overrightarrow{v_1}, \overrightarrow{v_2}, \dots, \overrightarrow{v_n}$

$$\underline{v} = a_1\overrightarrow{v_1} + a_2\overrightarrow{v_2} + \ \dots + a_n\overrightarrow{v_n}$$

Example:

Can RGB vectors $\{(255,0,0,), (0,255,0), (0,0,255)\}$ *create the color (150,150,150) Light Gray*

Answer: Yes! A set of vectors

$$0.59(255,0,0,) + 0.59(0,255,0) + 0.59(0,0,255) = 0.59\text{R} + 0.59\text{G} + 0.59\text{B}$$

Example:

Let $S = \{(1,0,0), (0,1,0), (0,0,1)\}$ be the set of standard unit vectors in R^3. Prove that S spans R^3

S spans R^3 means every vector of R^3 can be written as linear combination of these three vectors

i.e. $(a, b, c) = k_1\overrightarrow{v_1} + k_2\overrightarrow{v_2} + k_3\overrightarrow{v_3}$

$$(a, b, c) = k_1(1,0,0) + k_2(0,1,0) + k_3(0,0,1)$$
$$(a, b, c) = (k_1, 0,0) + (0, k_2, 0) + (0,0, k_3)$$
$$(a, b, c) = (k_1, k_2, k_3) \Rightarrow a = k_1, b = k_2, c = k_3$$
$$\Rightarrow (a, b, c) = a(1,0,0) + b(0,1,0) + c(0,0,1)$$

For example:

$$(1, 2, 3) = 1(1,0,0) + 2(0,1,0) + 3(0,0,1)$$

And

$$(15, -12, 7) = 15(1,0,0) - 12(0,1,0) + 7(0,0,1)$$

Please see the following MS Word file uploaded on portal for further practice problems

15. Linear Independence and Spanning


THANKS